

Climate Change & the Renewable Energy Transition

Science, Policy, and the Path to Net Zero

1. The Science of Climate Change

The Earth's climate is changing at an unprecedented rate, driven primarily by human activities that release greenhouse gases (GHGs) into the atmosphere. The Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (2021) concluded with unequivocal certainty that human influence has warmed the atmosphere, ocean, and land. Global surface temperature has already increased by approximately 1.1°C above pre-industrial levels (1850–1900 baseline), and current trajectories suggest we could breach the 1.5°C threshold by the early 2030s.

The primary greenhouse gases responsible for this warming are carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O). CO₂ concentrations in the atmosphere have risen from a pre-industrial level of approximately 280 parts per million (ppm) to over 420 ppm as of 2023—the highest level in at least 3 million years. The burning of fossil fuels (coal, oil, and natural gas) accounts for roughly 75% of global GHG emissions, with deforestation, agriculture, and industrial processes contributing the remainder. The enhanced greenhouse effect traps outgoing infrared radiation, causing the planet to warm in ways that disrupt weather patterns, sea levels, and ecosystems.

1.1 Observed and Projected Impacts

The consequences of climate change are already manifesting across the globe. Sea levels are rising at approximately 3.7 mm per year, driven by thermal expansion of ocean water and melting ice sheets in Greenland and Antarctica. Arctic sea ice extent has declined by roughly 13% per decade since satellite records began in 1979. Extreme weather events—heatwaves, droughts, heavy precipitation, and tropical cyclones—have become more frequent and intense. The 2021 Pacific Northwest heat dome killed over 1,400 people in Canada and the US, producing temperatures that scientists estimated would have been virtually impossible without climate change.

Projections for the end of the century depend heavily on emissions trajectories. Under a high-emissions scenario (SSP5-8.5), global temperatures could rise by 3.3–5.7°C, leading to catastrophic sea level rise, widespread ecosystem collapse, and severe food and water insecurity affecting billions of people. Under an aggressive mitigation scenario (SSP1-1.9), temperature rise could be limited to 1.0–1.8°C, significantly reducing these risks. The economic costs of unmitigated climate change are estimated in the trillions of dollars, with disproportionate impacts falling on low-income countries and vulnerable communities.

2. The Renewable Energy Revolution

Decarbonizing the global energy system is the single most important lever for addressing climate change. Electricity generation currently accounts for approximately 25% of global GHG emissions, and electrifying other sectors (transport, heating, industry) means the power sector must be decarbonized first and most aggressively. Fortunately, the renewable energy revolution is well underway, driven by dramatic cost reductions and growing policy support.

2.1 Solar and Wind Power

The cost of solar photovoltaic (PV) electricity has fallen by over 90% since 2010, making it the cheapest source of electricity in history in most parts of the world. Utility-scale solar now regularly achieves levelized costs of electricity (LCOE) below \$0.03/kWh in sun-rich regions—cheaper than operating existing coal or natural gas plants in many markets. Wind power has similarly seen dramatic cost reductions, with offshore wind costs falling by nearly 60% over the past decade. Global installed solar capacity surpassed 1 terawatt (TW) in 2022, and combined solar and wind capacity additions are breaking records annually.

2.2 Energy Storage

The intermittent nature of solar and wind power—they generate electricity only when the sun shines or wind blows—requires complementary energy storage solutions. Lithium-ion batteries, the same technology powering electric vehicles, have experienced cost declines of over 97% since the 1990s and are now being deployed at grid scale to shift solar generation from daytime to evening peak demand periods. Longer-duration storage technologies, including flow batteries, compressed air energy storage, green hydrogen, and pumped hydro, are needed to balance supply and demand across days, weeks, and seasons.

3. Policy Landscape and International Agreements

The Paris Agreement, adopted in 2015 by 196 nations, established the goal of limiting global warming to well below 2°C above pre-industrial levels and pursuing efforts to limit warming to 1.5°C. Under the agreement, countries submit Nationally Determined Contributions (NDCs) outlining their emission reduction targets. However, a 2022 UN analysis found that current NDCs, even if fully implemented, would result in warming of approximately 2.5°C by 2100—far exceeding the 1.5°C goal.

The US Inflation Reduction Act (IRA) of 2022 represents the largest climate investment in US history, allocating approximately \$369 billion for clean energy and climate programs over a decade. The European Union's Green Deal aims to make Europe the first climate-neutral continent by 2050, supported by the Fit for 55 package of legislative reforms targeting a 55% reduction in net GHG emissions by 2030 relative to 1990 levels. China, the world's largest emitter, has pledged to peak emissions before 2030 and achieve carbon neutrality before 2060, while massively scaling its renewable energy capacity.

4. The Just Transition

Transitioning to a clean energy economy will create millions of new jobs in manufacturing, installation, and maintenance of renewable energy systems, electric vehicles, and energy efficiency retrofits. However, it will also displace workers in fossil fuel industries, particularly in coal-dependent communities. A just transition requires targeted policies to support affected workers through retraining programs, economic diversification investments, and social safety nets. Indigenous communities and low-income populations, who often bear a disproportionate burden of both climate impacts and fossil fuel pollution, must be centered in climate policy design.

5. Conclusion

The science is clear: rapid, deep, and sustained reductions in greenhouse gas emissions are necessary to avoid the most catastrophic consequences of climate change. The good news is that renewable energy technologies have reached cost parity with or undercut fossil fuels in most markets, and the economic case for climate action has never been stronger. The challenge is not technological but political, institutional, and social—mobilizing the will to redirect investment, reform policies, and transform energy systems at the speed and scale that science demands. The decisions made in this decade will determine the climate trajectory for centuries to come.

Table 2: Renewable Energy Cost Trends (2010–2023)

Technology	2010 LCOE (\$/MWh)	2023 LCOE (\$/MWh)	Cost Reduction
Utility Solar PV	\$378	\$29	~92%
Onshore Wind	\$135	\$33	~76%
Offshore Wind	\$188	\$81	~57%
Li-ion Battery Storage	\$1,200/kWh	\$139/kWh	~88%
Coal (new plant)	\$111	\$108	~3%
Natural Gas (CCGT)	\$83	\$60	~28%